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Rotation-Invariant Neural Networks

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Introduction

Images surround us everywhere: from smartphone photos, satellite imagery, and city surveillance, through product catalogs and quality control on production lines, to driver-assistance systems. Although modern image recognition models perform remarkably well, in practice they can be sensitive to seemingly minor changes such as rotating an object by a few degrees or slightly tilting the camera. What is natural and immediately recognizable to a human (a road sign at an angle, a digit rotated on paper) can be problematic for a classical convolutional neural network. The core difficulty is the lack of natural invariance to rotations: standard CNNs are "by definition" better at handling translations than rotations [1], [2].

In recent years, several approaches have been proposed to close this gap. One is augmenting the data with rotated examples – which improves robustness but increases training time and does not guarantee generalization to all angles. Another approach is designing architectures with built-in geometry: group-equivariant networks (G-CNN, E(2)-equivariant) [3], [4], cyclic networks (CyCNN; in particular **CyVGG** and **CyResNet**) operating across multiple orientations, and transformations into polar coordinates (linear-polar and log-polar), which "straighten" rotations into translations. The common goal is for the model to recognize "the same" object regardless of orientation, without aggressive data duplication.

This thesis focuses on the practical evaluation of these approaches. Datasets were prepared including handwritten digits, road signs (in color and grayscale), and synthetic 3D objects projected onto 2D (LEGO bricks), which were then extended with controlled rotations. Selected rotation-invariant architectures and their baseline variants were implemented and compared in **PyTorch** [5], measuring the impact of transformations (linear-polar vs. log-polar), architectural choices, and angular ranges on prediction quality. The experiments were carried out on **NVIDIA GeForce RTX 3070 Ti 8 GB** and **RTX 3060 12 GB** GPUs, which significantly reduced training times and enabled a broad experimental survey; the runtime environment was standardized using **Docker** for reproducibility.

The aim of this work is not only to show that it is possible to achieve rotation invariance, but more importantly to indicate **when** and **at what cost** it can be achieved, which techniques yield the greatest improvement over classical CNNs, how they affect stability and training speed, and which configurations are the most practical in real-world applications (OCR, traffic sign recognition, analysis of technical objects). The following chapters present the theoretical background, datasets and augmentation strategies, architectures, experimental environment, evaluation protocols, results, and conclusions.

References

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